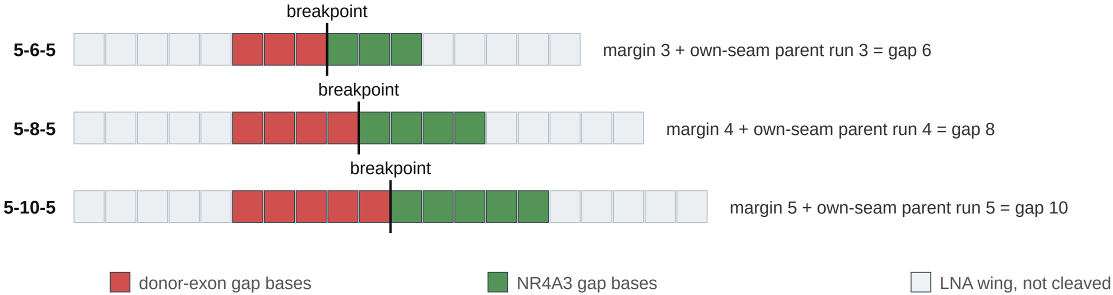
A longer catalytic gap buys junction-unique bases and concedes gap DNA at the design's own seam, one for one

This ordinate is the design's own seam arithmetic, not the mature-parent search Tables 2, 3, 5 and 7 print under “longest parent duplex through the gap”. That search FALLS as the gap lengthens (181 of 190 at 5-6-5; 130 of 266 at 5-8-5; 87 of 342 at 5-10-5) and is FLAT at the ten-base-pair criterion (87 of 190 at 5-6-5; 88 of 266 at 5-8-5; 87 of 342 at 5-10-5). Only the quantity drawn here rises.

A · the catalytic gap is tiled by the two counts

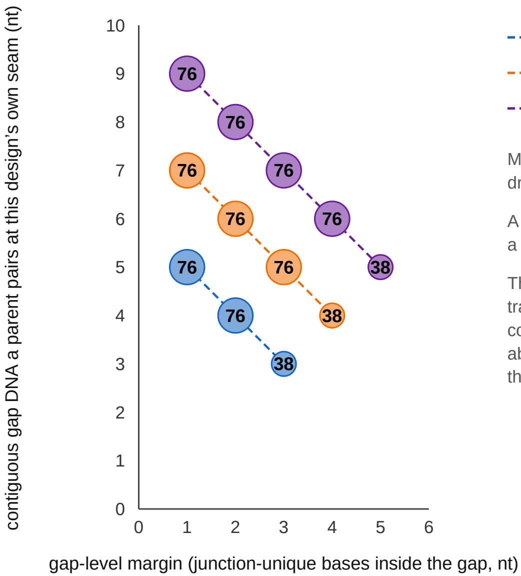
The best-margin design at EWSR1 e12 :: NR4A3 e3, at each geometry. Wings fixed at five nucleotides, so only the gap changes. The register drawn divides its gap evenly at every geometry, which is the highest margin the geometry allows and not the common case: 38 of 190 at 5-6-5; 38 of 266 at 5-8-5; 38 of 342 at 5-10-5. Panel B counts every register there is.



The same three molecules' mature-parent duplex through the whole gap — the column the tables print — reads 8 bp (TFG) at 5-6-5, none at 5-8-5, none at 5-10-5. It is a search over six wild-type parent transcripts, not this arithmetic, and on this molecule the two disagree.

B · every design in all three geometries, with no scatter to fit

798 designs over 38 junctions. Both axes are drawn at 32 px per nucleotide, so the identity lines really are at 45 degrees.



Marker area is the number of designs at that point and the label is that count; the three lines are drawn from the identity, not fitted.

A geometry's ceiling on margin is half its gap rounded down; clearing it means a longer gap, and a longer gap raises the own-seam parent-paired run at every register.

The identity holds for every design individually, not on average. A longer gap also buys a quieter transcriptome: over the 6 junctions every geometry was screened at, the median near-match count falls 86.5 to 15 to 0 across 5-6-5, 5-8-5, 5-10-5. Part of that fall is arithmetic — at a fixed absolute mismatch budget a longer probe cannot hit more loci than its own sub-windows — and this figure is what the rest of it costs.